Eigenvalues & Eigenvectors

Geometric Algorithms

CAS CS 132

Practice Problem

rk A + nullity A = n
dim(Col(A)) + dim(Nul(A)) = n
(for m x n matrix)

Suppose A *is a* 234×300 *matrix. What is the smallest possible value for* $\dim(\text{Nul}(A))$*? What is the largest possible value?*

What is the smallest possible value for $\text{rank}(A)$*? What is the largest possible value?*

Answer

A is 234×300

234×300

$$\begin{bmatrix} 1 & * & 0 & * & 0 & * & * \\ & & 1 & * & 0 & * & * \\ & & & & 1 & * & * \\ 0 & 0 & 0 & \cdots & & & 0 \end{bmatrix}$$

pivot columns/pos'ns in rref(A)

columns in rref(A) (free variables)

$\dim(\mathrm{Col}(A)) + \dim(\mathrm{Nul}(A)) = 300$

\# of pivot columns $=$ \# of non-pivots

$66 \leq \dim(\mathrm{Nul}(A)) \leq 300$

achieved by $A = \begin{bmatrix} 0 & \cdots \\ \vdots & \ddots \end{bmatrix}$

No more than 234 pivot positions

$\dim(\mathrm{Nul}(A)) \geq 300 - 234 = 66$

$0 \leq \mathrm{rk}\ A = \dim(\mathrm{Col}(A)) \leq 234$

achieved by 0 matrix

$\mathrm{rk}\ A \leq \min(m, n)$

Objectives

- Motivate and introduce the fundamental notion of eigenvalues and eigenvectors
- Determine how to **verify** eigenvalues and eigenvectors
- Look at the **subspace** generated by eigenvectors
- Apply the study of eigenvectors to **dynamical linear systems**

Keyword

Eigenvalues

Eigenvectors

Null Space

Eigenspace

Linear Dynamical Systems

Closed-Form Solutions

Motivation

shear: $\begin{bmatrix} 1 & 0.01 \\ 0 & 1 \end{bmatrix}$

scale: $\begin{bmatrix} 1.02 & 0 \\ 0 & 0.98 \end{bmatrix}$

demo

How can matrices transform* vectors?

- rotations
- projections
- shearing
- reflection
- scaling/stretching
- ... **Today's focus**

} All matrices do some combination of these things

* square matrices

What's special about scaling?

We don't need a matrix, it's just $\mathbf{x} \mapsto c\mathbf{x}$

If $A\mathbf{v} = c\mathbf{v}$ then it's "easy to describe" what A does to $\mathbf{v}$

Eigenvectors (Informal)

eigenvalue

$$A\mathbf{v} = \lambda\mathbf{v}$$

eigenvector

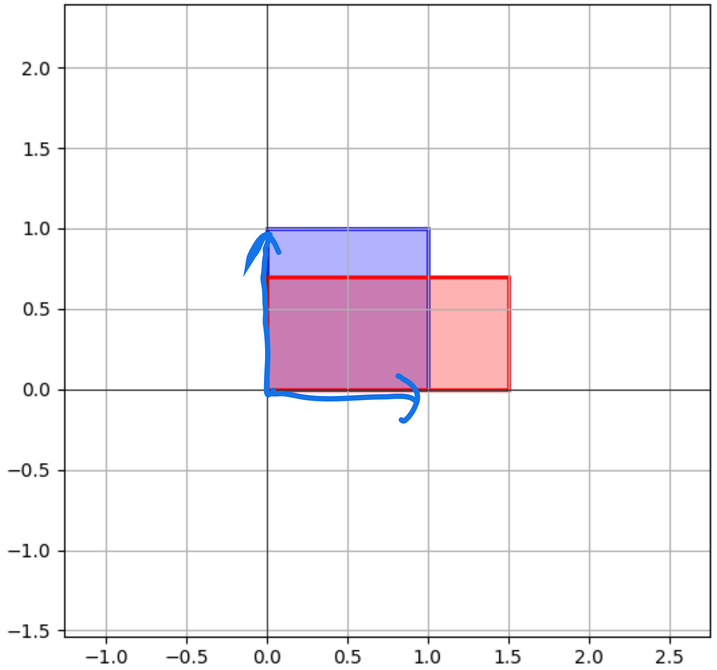
Eigenvectors of A are stretched by A without changing their direction

The amount they are stretched is called the **eigenvalue**

Example: Unequal Scaling

$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix}$$

It's "easy to describe" how unequal scaling transforms vectors. *It transforms each entry individually and then combines them*



Eigenbases (Informal)

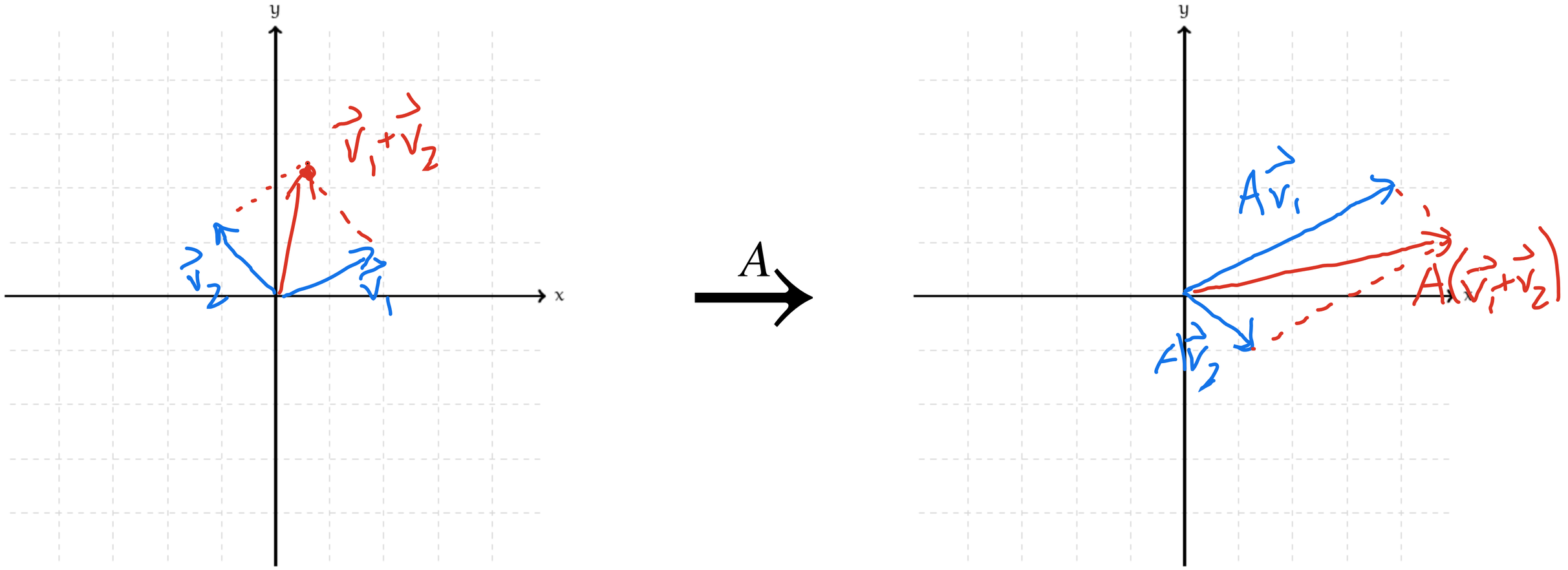
$\lambda_1, \lambda_2, \lambda_3$

Suppose $\mathbf{v} = 2\mathbf{b}_1 - \mathbf{b}_2 - 5\mathbf{b}_3$ and $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ are *eigenvectors of* A

It's "easy to describe" how A transforms $\mathbf{v}$:

$$A\vec{v} = 2A\vec{b}_1 - A\vec{b}_2 - 5A\vec{b}_3 = 2\lambda_1\vec{b}_1 - \lambda_2\vec{b}_2 - 5\lambda_3\vec{b}_3$$

Eigenbases (Pictorially)

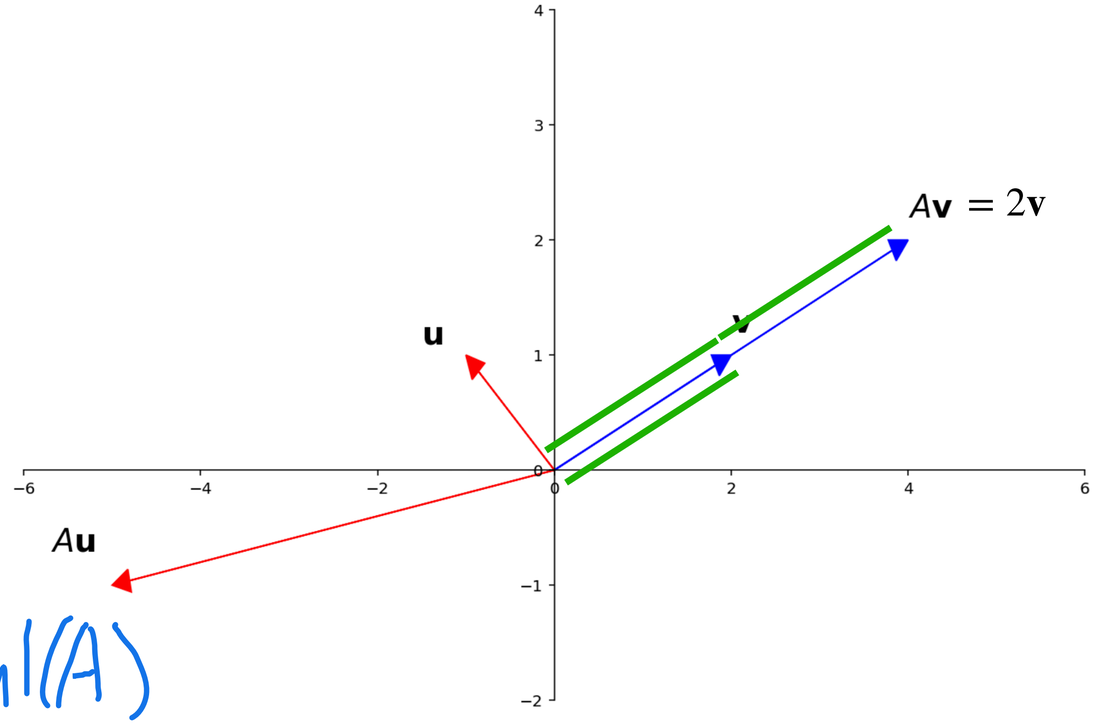


Eigenvalues and Eigenvectors

Formal Definition

$$A\vec{0} = \vec{0} = (0)(\vec{0})$$

$$A\vec{v} = 0(\vec{v}) = \vec{0} \iff \vec{v} \in Nul(A)$$



Def. A *nonzero* vector $\mathbf{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ are an **eigenvector and eigenvalue** for a $A \in \mathbb{R}^{n \times n}$ if $A\mathbf{v} = \lambda\mathbf{v}$

We will say that $\mathbf{v}$ is an eigenvector *of/for* the eigenvalue λ, and that λ is the eigenvalue *of/corresponding to* $\mathbf{v}$

Note. Eigenvectors *must* be nonzero, but 0 can be an eigenvalue

What if 0 is an eigenvalue?

» $\mathbf{v} \in \text{Nul}(A)$

» $\mathbf{v}$ is a nontrivial solution to $A\mathbf{v} = \mathbf{0}$

Extending the IMT (Again)

Thm. A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if it *does not* have 0 as an eigenvalue

To reiterate, having eigenvalue 0 is equivalent to:

» $\text{Nul}(A) = \{\vec{0}\}$

» $A\mathbf{x} = \mathbf{0}$ has no nontrivial solutions

» the columns of A are linearly dependent

» $\text{Col}(A) \neq \mathbb{R}^n$

» $\vec{0}$ is not an eigenvalue

Example: Unequal Scaling

$$A = \begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix}$$

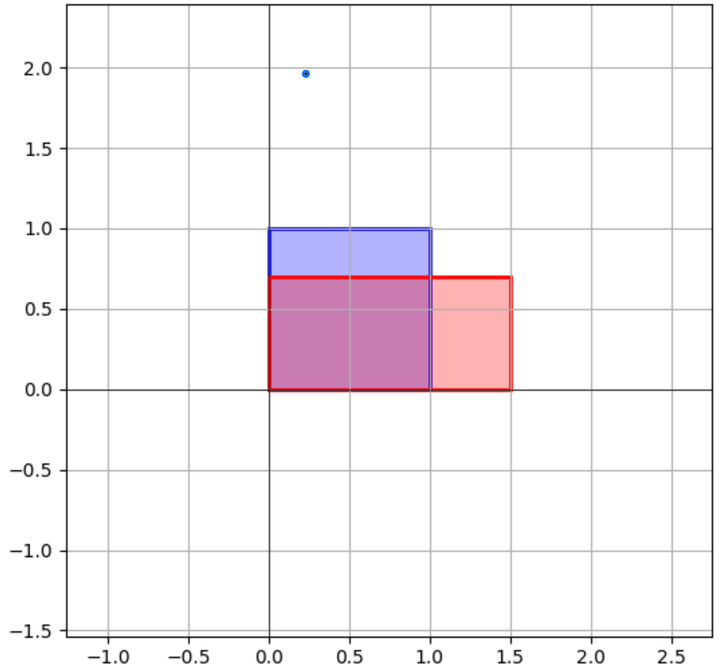
λ_1?

$$A\vec{e}_1 = \begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 0 \end{bmatrix} = \lambda_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (1.5)\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

λ_2?

$$A\vec{e}_2 = \begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.7 \end{bmatrix} = \lambda_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (0.7)\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 0.7 \end{bmatrix}$$



Example: Shearing

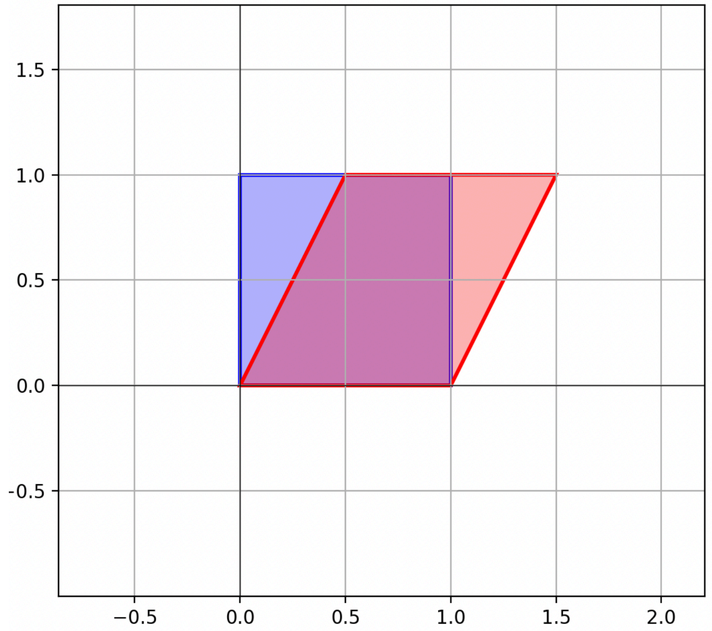
$$\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$$

$\vec{e}_1$ $\quad\quad$ λ_1

$$\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (1) \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix}$$



Example

$$A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix} \quad \mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{v} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$

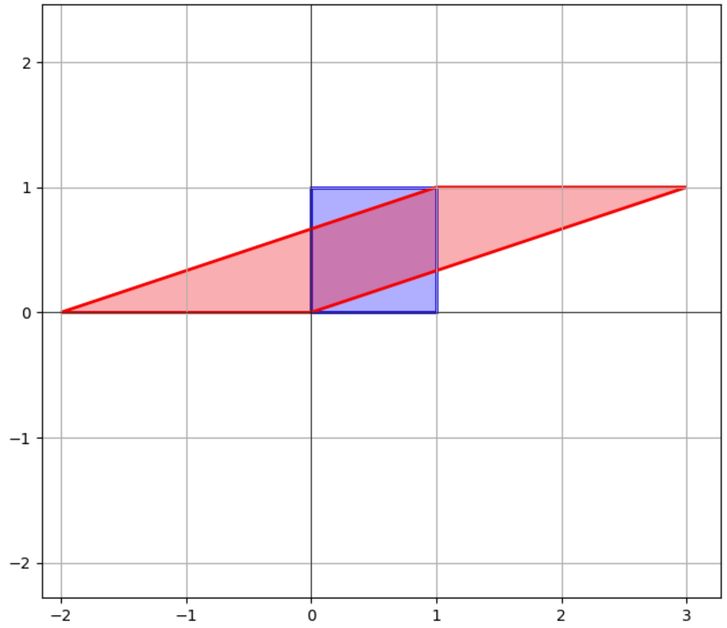
$$A\vec{u} = \begin{bmatrix} 3-2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1)\vec{u}$$

$\lambda_u = 1$

$$A\vec{v} = \begin{bmatrix} 3-1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} = (2)\begin{bmatrix} 1 \\ 0.5 \end{bmatrix} = 2\vec{v}$$

$\lambda_v = 2$

$\begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$



How do we verify eigenvalues and eigenvectors?

Verifying Eigenvectors (practice for later)

Q. *Determine if* $\begin{bmatrix} 6 \\ -5 \end{bmatrix}$ *or* $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ *are eigenvectors of* $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ *and determine their eigenvalues*

A. Easy, work out the mv multiplication

Verifying Eigenvalues

This is harder...

Q. Show that 7 is an eigenvalue of $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$

What vector do we check?

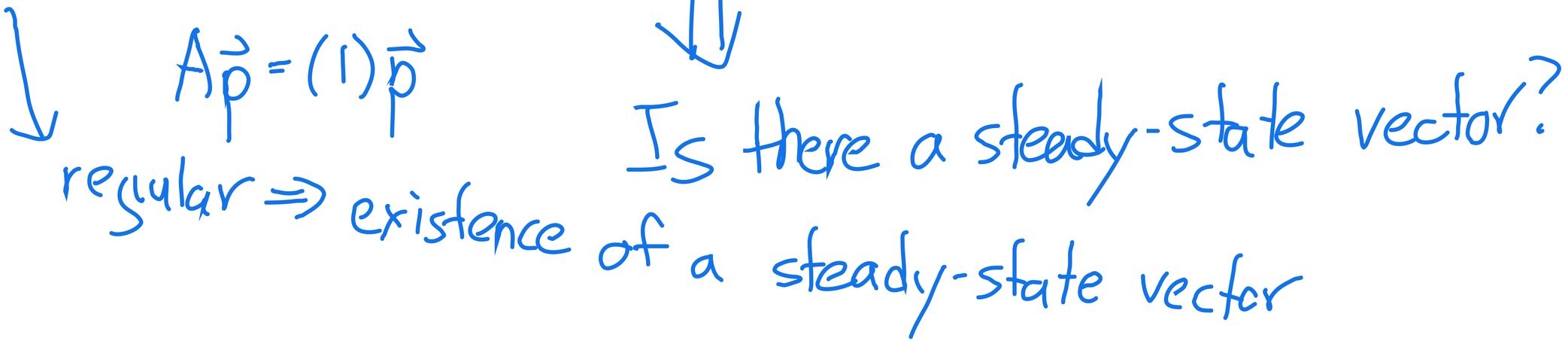
Before we go over how to do this...

Verifying Eigenvalues (Warm Up)

Q. Verify that 1 is an eigenvalue of

$$\begin{bmatrix} 0.1 & 0.7 \\ 0.9 & 0.3 \end{bmatrix}$$ (*Hint.* Recall Markov Chains)

A.



Steady-States and Eigenvectors

The Point: *Steady-state distributions are eigenvectors with eigenvalue* 1

$A\vec{v} = \vec{v} \Leftrightarrow (A - I)\vec{v} = 0$

$\mathbf{v}$ is a steady-state vector $\equiv$

$$\mathbf{v} \in \text{Nul}(A - I)^*$$

*It must also be a probability vector

Verifying Eigenvalues

Q. Show that λ is an eigenvalue of A

A. is there $\vec{v} \neq 0$ s.t. $A\vec{v} = \lambda\vec{v} \Leftrightarrow A\vec{v} - \lambda\vec{v} = (A - \lambda I)\vec{v} = 0$

$\Leftrightarrow \vec{v} \in Nul(A - \lambda I)$

$\mathbf{v}$ is an eigenvector for λ $\equiv$

$$\mathbf{v} \in \mathbf{Nul}(A - \lambda I)$$

(& $\vec{v} \neq 0$)

Verifying Eigenvalues

Q. Show that 7 is an eigenvalue of $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$

A. $A - 7I = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \Rightarrow$ $V_1 = V_2$, V_2 free

$\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$ a choice

$(A - 7I)\vec{v} = 0$

A nice check $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.5+3 \\ 2.5+1 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 3.5 \end{bmatrix} = 7\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$

Some Practice Problems Now

Verifying Eigenvectors & Eigenvalues

Q1. *Determine if* $\begin{bmatrix} 6 \\ -5 \end{bmatrix}$ *or* $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ *are eigenvectors of*

$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ *and determine their eigenvalues*

Q2. Verify that 2 is an eigenvalue of:

$$\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$$

Verifying Eigenvectors

Q. *Determine if* $\begin{bmatrix} 6 \\ -5 \end{bmatrix}$ *or* $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ *are eigenvectors of*

$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ *and determine their eigenvalues*

$\vec{u}$

$\vec{v}$

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}\begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} 6-30 \\ 30-10 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = (-4)\begin{bmatrix} 6 \\ -5 \end{bmatrix}$$

✓

$= \lambda u$

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}\begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 3-12 \\ 15-4 \end{bmatrix} = \begin{bmatrix} -9 \\ 11 \end{bmatrix} \neq c\begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

✗

Problem

$$\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 4-2 \\ 2+2 \\ 2-2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

Verify that 2 *is an eigenvalue of* $\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$

$A\vec{v} = 2\vec{v} \Leftrightarrow (A - 2I)\vec{v} = 0$

$$\begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & -\frac{1}{2} & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{array}{l} v_1 = \frac{1}{2}v_2 - 3v_3 \\ v_2 \text{ free} \\ v_3 \text{ free} \end{array}$$

$v_2 = 2$, $v_3 = 0$ $\Rightarrow \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ a choice of eigenvector

Answer

$$\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$$

How many eigenvectors can a matrix have?

Linear Independence of Eigenvectors

$\lambda_1 \neq \cdots \neq \lambda_k$

Thm. If $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are eigenvectors for distinct eigenvalues, then they are linearly independent

So an $n \times n$ matrix can have at most n distinct eigenvalues

Eigenspace

Fact. The set of eigenvectors for a eigenvalue λ of $A \in \mathbb{R}^{n \times n}$ form a subspace of $\mathbb{R}^n$

Closure under add'n: $A\vec{v} = \lambda\vec{v}$, $A\vec{w} = \lambda\vec{w}$

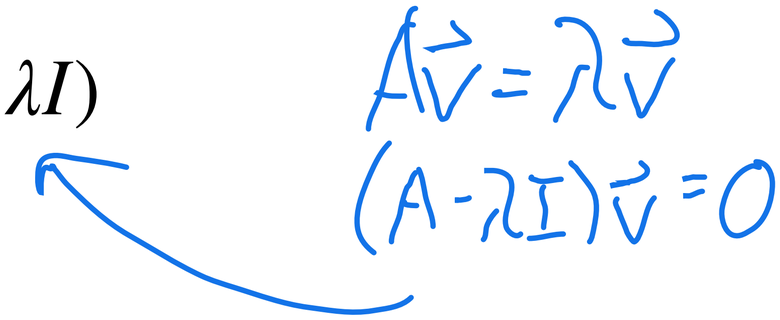
$A(\vec{v}+\vec{w}) = A\vec{v} + A\vec{w} = \lambda\vec{v} + \lambda\vec{w} = \lambda(\vec{v}+\vec{w})$

Closure under scaling: exercise

Eigenspace

Def. The set of eigenvectors for a eigenvalue λ of A is called the **eigenspace** of A for λ

It is the same as $\text{Nul}(A - \lambda I)$



How To: Basis of an Eigenspace

Q. Find a basis for the eigenspace of A corresponding to λ

A. Find a basis for $\text{Nul}(A - \lambda I)$

We know how to do this

find gen. sol'n

Example

$$A = \begin{bmatrix} -2 & 0 & 3 \\ 1 & 1 & -1 \\ -4 & 0 & 5 \end{bmatrix}$$

Determine a basis for the eigenspace corresponding to the eigenvalue 1

$$A - I = \begin{bmatrix} -3 & 0 & 3 \\ 1 & 0 & -1 \\ -4 & 0 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{array}{l} v_1 = v_3 \\ v_2 \text{ free } (v_2 = v_2) \\ v_3 \text{ free } (v_3 = v_3) \end{array}$$

$(A-I)\vec{v} = \vec{0}$

$$\vec{v} = v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + v_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

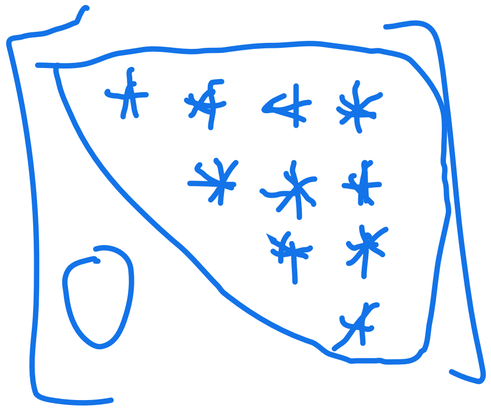
$\{\ ,\ \}$ a basis for $\text{Nul}(A-I)$

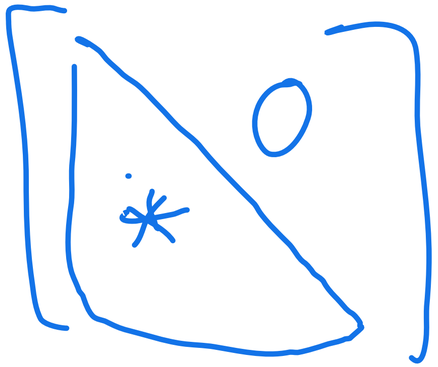
How do we find eigenvalues?

We'll cover this next time...

Eigenvalues of Triangular Matrices

Thm. The eigenvalues of a triangular matrix are its entries along the diagonal





Example

$$A = \begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

$\text{Nul}(A - \lambda_2 I)$

$v_1 = -10v_3$, $v_2 = 3v_3$, v_3 free $\Rightarrow$ $v_3 = 1$: $\begin{bmatrix} -10 \\ 3 \\ 1 \end{bmatrix} = \vec{v}_3$

Determine the eigenvectors and values of the above matrix

$\lambda_1 = 3$, $\lambda_2 = 0$, $\lambda_3 = 2$ eigenvalues

$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ a reasonable choice of eigenvector

$\vec{v}_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ a choice of eigenvector

$$A - 2I = \begin{bmatrix} 1 & 6 & -8 \\ 0 & -2 & 6 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 6 & -8 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

Linear Dynamical Systems

Recall: Linear Dynamical Systems

Def. A **(discrete time) linear dynamical system** is a described a $n \times n$ matrix A. It's **evolution function** is the matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$

The **possible states** of the system are vectors in $\mathbb{R}^n$

Given an **initial state vector** $\mathbf{v}_0$, we can determine the **state vector** of the system after i time steps:

$$\mathbf{v}_i = A\mathbf{v}_{i-1}$$ *(linear recurrence relation)*

Recall: State Vectors

$$
\begin{aligned}
&\mathbf{v}_0 \\
&\mathbf{v}_1 = A\mathbf{v}_0 \\
&\mathbf{v}_2 = A\mathbf{v}_1 = A(A\mathbf{v}_0) \\
&\mathbf{v}_3 = A\mathbf{v}_2 = A(AA\mathbf{v}_0) \\
&\mathbf{v}_4 = A\mathbf{v}_3 = A(AAA\mathbf{v}_0) \\
&\mathbf{v}_5 = A\mathbf{v}_4 = A(AAAA\mathbf{v}_0) \\
&\quad\vdots
\end{aligned}
$$

The **state vector** $\mathbf{v}_k$ is the state of the system after k time steps

The Issue

The eq. $\mathbf{v}_k = A^k \mathbf{v}_0$ doesn't tell us much about $\mathbf{v}_k$

It's defined in terms of A itself, which doesn't say much about how the system behaves

It's also difficult computationally because matrix multiplication is expensive

(Closed-Form) Solutions

A **(closed-form) solution** of a linear recurrence relation $\mathbf{v}_{i+1} = A\mathbf{v}_i$ is an expression for $\mathbf{v}_k$ that does *not* contain A^k or previously defined terms

In other word, it doesn't depend on A^k and isn't **recursive**

(Non-)Example

$$\mathbf{v}_k = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{v}_{k-1} \qquad \mathbf{v}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Solutions with Eigenvectors as Initial States

It's easy to give a closed-form solution if the initial state is an eigenvector

No dependence on A^k or $\mathbf{v}_{k-1}$

$$\mathbf{v}_k = A^k \mathbf{v}_0 = \lambda^k \mathbf{v}_0$$

The Key Point. *This is still true of linear combinations of eigenvectors*

Solutions in terms of eigenvectors

Let's simplify $A^k\mathbf{v}$, given we have eigenvectors $\mathbf{b}_1, \mathbf{b}_2$ for A which span all of $\mathbb{R}^2$:

Eigenvectors and Growth in the Limit

Thm. For a linear dynamical system A with initial state $\mathbf{v}_0$, if $\mathbf{v}_0$ can be written in terms of eigenvectors $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k$ of A with eigenvalues $\lambda_1 > \lambda_2 \ldots \geq \lambda_k$ then $\mathbf{v}_k \sim \lambda_1^k c_1 \mathbf{b}_1$ for some constant c_1

In other words, in the long term, the system grows exponentially in the largest eigenvalue

Eigenbases

Def. An **eigenbasis** of $\mathbb{R}^n$ for $A \in \mathbb{R}^{n \times n}$ is a basis of $\mathbb{R}^n$ made up entirely of eigenvectors of A

*We can represent vectors as **unique** linear combinations of eigenvectors*

Not all matrices have eigenbases

Eigenbases and Growth in the Limit

Thm. For a linear dynamical system A with initial state $\mathbf{v}_0$, if A has an eigenbasis $\mathbf{b}_1, \ldots, \mathbf{b}_k$, then

$$\mathbf{v}_k \sim \lambda_1^k c_1 \mathbf{b}_1$$

for some constant c_1, where where λ_1 is the **largest eigenvalue of** A **and** $\mathbf{b}_1$ **is its eigenvalue**

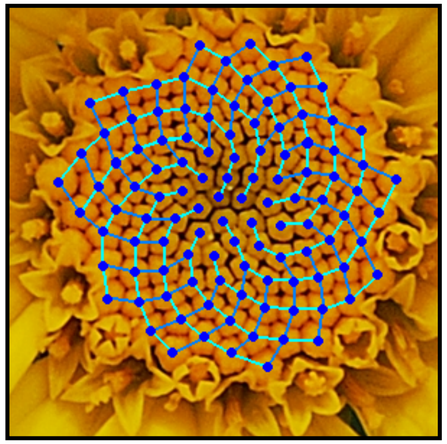
The largest eigenvalue describes the long-term exponential behavior of the system

Example: Golden Ratio

A Special Linear Dynamical System

$$\mathbf{v}_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{v}_k \qquad \mathbf{v}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Fibonacci Numbers



$$F_0 = 0$$

$$F_1 = 1$$

$$F_k = F_{k-1} + F_{k-2}$$

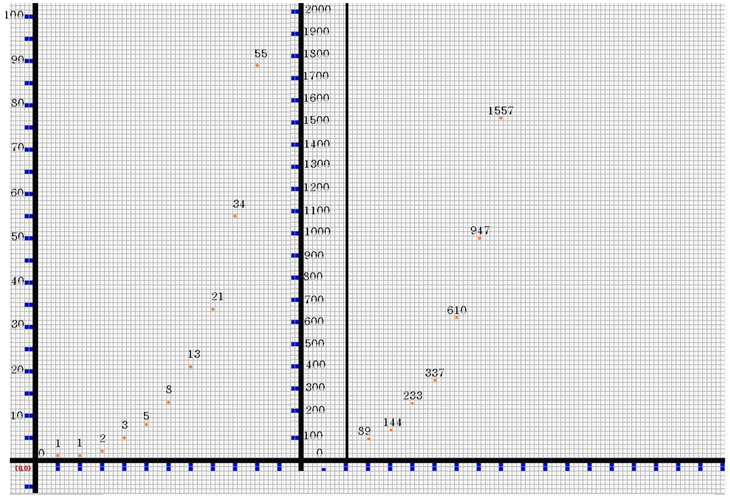
```
define fib(n):
  curr, next ← 0, 1
  repeat n times:
    curr, next ← next, curr + next
  return curr
```

The Fibonacci numbers are defined in terms of a recurrence relation

https://commons.wikimedia.org/wiki/File:FibonacciChamomile.PNG

Golden Ratio

$$\varphi = \frac{1+\sqrt{5}}{2}$$



The "long term behavior" is the Fibonacci sequence is defined by the golden ratio: $F_k \sim \varphi^k$

https://commons.wikimedia.org/wiki/File:Fibonacci_Sequence_Graph.PNG